INCH-POUND
MIL-DTL-18014C
8 May 2019
SUPERSEDING
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15 March 1967

DETAIL SPECIFICATION

GRATINGS, METAL, BAR TYPE FLOORING, NAVAL SHIPBOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers open, bar type metal gratings (flooring) for naval shipboard use.

1.2 Classification. Gratings are of the following types, classes, and styles, as specified (see 6.2).

a. Type A – Steel, heavy duty, galvanized
   (1) Class A – Lightweight
   (2) Class B – Heavyweight

b. Type B – Aluminum alloy, light duty
   (1) Class A – Lightweight
      (a) Style 1 – Riveted
      (b) Style 2 – Interlocked

1.3 Part or identifying number (PIN). PINs to be used for gratings acquired to this specification are created as follows:

M	**18014**	**-**	**X**	**-**	**X**	**-**	**X**
Prefix for Military Specification	Specification Number		Type (A or B)		Class (A or B)		Style (Class A only) (1 or 2)
Examples: M18014-A-B, M18014-B-A-2							

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUE (ANSI)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection of Attributes

(Copies of this document are available online at http://webstore.ansi.org/.)

ASTM INTERNATIONAL

ASTM A105/105M	-	Standard Specification for Carbon Steel Forgings for Piping Applications
ASTM A109/109M	-	Standard Specification for Steel, Strip, Carbon (0.25 Maximum Percent), Cold-Rolled
ASTM A153/A153M	-	Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A240/240M	-	Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A666	-	Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate, and Flat Bar
ASTM A693	-	Standard Specification for Precipitation-Hardening Stainless and Heat-Resisting Steel Plate, Sheet, and Strip
ASTM B209	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B211	-	Standard Specification for Aluminum and Aluminum-Alloy Rolled or Cold Finished Bar, Rod, and Wire
ASTM B221	-	Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B241/241M	-	Standard Specification for Aluminum and Aluminum-Alloy Seamless Pipe and Seamless Extruded Tube

(Copies of these documents are available online at www.astm.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material.

3.2.1 Type A. Type A gratings shall be made of steel in accordance with ASTM A240/240M, ASTM A666, ASTM A693, ASTM A105/105M, or ASTM A109/109M.

3.2.2 Type B. Type B gratings shall be made of aluminum alloy in accordance with ASTM B241/B241M, ASTM B221, ASTM B211, or ASTM B209.

3.2.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 General.

3.3.1 Size and shape. The grating panel sizes shall be as specified (see 6.2). Where nonrectangular shapes, or cutouts in rectangular panels are required, they shall be as shown on the drawings as specified (see 6.2). In general, the edges of nonrectangular shapes and cutouts, where more than four bearing bars are included, shall be finished with banding bars of the same depth as the bearing bars. The banding bars shall be welded to the bearing bars of the grating at least every other bearing bar and in every corner. Intervening bearing bars shall be tack welded. Where banding bars are for the general appearance or finish only, welding to every fourth bearing bar is permitted. The ends of 6000 series aluminum grating panels shall be finished or banded by means of spacers or bridges to preclude welding. The 5000 series aluminum may be welded.

3.3.2 Clamps and fastenings. Clamps or other fastenings shall be provided when specified (see 6.2). Such clamps or fastenings shall be of the same material, be finished in the same manner as the gratings, and be of such construction as to provide convenient means for holding, securing, removing, and replacing the gratings to the supporting structure. Clamps or fastenings shall not extend above the surface of the gratings. 6000 series aluminum shall not be welded.

3.3.3 Deflection and set. The thickness and spacing of all bearing bars, and the depth, thickness, and spacing of all cross or reticuline bars shall be designed to support the loads specified in 3.6 without deflection due to bending and without lateral displacement, warping, or distortion of any of the parts greater than 0.2 inch and without permanent set greater than 0.008 inch (see 4.6.2).

3.4 Construction. The weight of gratings (pounds per square foot) shall not exceed the limits of table I.

3.4.1 Reticular. Gratings shall be reticular, flush topped (see 3.4.1.1), and constructed of straight parallel bearing bars, placed edgewise and joined by reticuline bars or cross bars. For steel gratings, reticuline bars or cross bars shall be secured to the bearing bars by riveting, welding, or interlocking joint construction (see table I); for aluminum gratings, style 1, securing shall be by riveting only (see table I); for aluminum gratings, style 2, securing shall be by interlocking only (see table I).

3.4.1.1 Cross bars. Gratings having cross bars which are not flush with the top of the bearing bars are acceptable, provided the top edges of the bearing bars are serrated to give the necessary skid resistance in the direction of the bearing bars.

3.4.2 Spacing of bearing bars. The spacing between the bearing bars shall not exceed 1⅞ inches from center to center of adjacent bars. For gratings of the type where the reticuline bars are not parallel cross bars and bent or crimped to contact the latter bars at intervals at which they are fastened to the bearing bars by rivets or welding, the spacing of rivets or welds shall not exceed 7 inches. For gratings of the type where the reticuline bars are parallel cross bars secured to the bearing bars by clamping, welding, or interlocking joint construction, the spacing of the cross bars shall not exceed 4 inches.

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3.4.3 Tolerances. Allowable tolerances shall be as follows:

a. Length of bearing bars – ±1/16 inch in 6 feet.

b. Depth of bearing bars – ±1/32 inch (when measured from bottom of serrations when present).

c. Width of grating panels – ±1/8 inch.

A tolerance of plus 10 percent is permitted on the maximum weight of any individual grating when banding bars or fasteners, or both, form a part of the grating.

3.5 Dimensions. The depth of the bearing bars shall be 1 inch, (included depth of serrations, when present) within the tolerance specified in 3.4.3. The depth of cross bars or reticuline bars shall not exceed 1 inch.

3.5.1 Class A. The thickness of bearing bars for class A gratings shall be not less than 1/8 inch.

3.5.2 Class B. The thickness of bearing bars for class B gratings shall be not less than 3/16 inch.

3.6 Load. The load requirements of the gratings shall be as shown in table I.

TABLE I. Load requirements.

Type, class, and style	Design loading per square foot (pounds)	Test load concentrated midway between supports on a 2-foot long 3-foot span (see 4.6) (pounds)	Maximum allowed weight per square foot, finished unbanded, including painting or zinc-coating (galvanized) (pounds)
Type A, class A – welded, interlocked or riveted	300	900	5.00
Type A, class B – welded, interlocked or riveted	300	900	8.45
Type B, class A, style 1 or 2	175	525	3.50

3.7 Zinc-coating (galvanizing). Unless otherwise specified (see 6.2), steel gratings shall be completely and evenly zinc-coated (galvanized) by the hot-dip process in accordance with ASTM A153 after fabrication. The coating shall not flake off when tested as specified in 4.6.

3.8 Workmanship. Gratings shall be accurately fabricated and free from warps, twists, or other defects affecting the appearance or serviceability. The bars from which the gratings are fabricated shall be sound, of uniform quality and condition, and free from harmful nonmetallic inclusions, segregations, pipes, laps, cracks, twists, seams, damaged ends, and other injurious defects.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on gratings when a first article sample is required (see 3.1). This inspection shall include the examination of 4.5 and the test of 4.6.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.5.

4.4 Sampling.

4.4.1 Lot. For purposes of sampling, a lot shall consist of all gratings of the same type, class, size, and style offered for delivery at the same time.

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4.4.1.1 Sampling for examination. A random sample of grates shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and Acceptance Quality Limit (AQL) as specified (see 6.2).

4.5 Examination. Each sample grating selected in accordance with 4.4.1.1 shall be examined to determine conformance with the material requirements (see 3.1), workmanship (see 3.8), and the weight and dimensional requirements of this specification (see 3.3, 3.3.3, and table I).

4.6 Load test. Each sample item shall be subjected to a load test. Each item shall be supported by two supports with bearing width not to exceed 1 inch. The supports shall be placed 3 feet apart. The gratings shall be tested with the load specified for the type and class in accordance with table I, suspended from a bar placed midway between the supports. Deflection and permanent set shall be determined by gages.

4.6.1 Deflection criteria. The maximum deflection measured at the center or the grating shall not exceed 0.2 inch. There shall be no visible lateral displacement, warping, or distortion (see 3.3.3).

4.6.2 Permanent set criteria. The maximum permanent set measured at the center of the grating shall not exceed 0.008 inch (see 3.3.3).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Gratings manufactured to this specification are intended for use in a maritime environment with exposure to salt and heavy loads. Gratings are expected to function for the life of a naval ship and withstand shock loadings associated with a military environment. When other than bar type gratings are required, MIL-DTL-18015 should be used. When the intended use does not restrict the type of grating, either MIL-DTL-18015 or this specification should be used. Current stock of gratings may be used until depleted; future requisitions for gratings should refer to the current version of this specification. Equipment manufactured in accordance with this document may utilize new technology provided form, fit, and function are maintained as approved by the Technical Authority.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class of gratings required (see 1.2).
c. Whether first article test is required (see 3.1).
d. Size of grating panels and drawings required (see 3.3.1).
e. Whether clamps or fastenings are required (see 3.3.2).
f. When steel gratings other than zinc-coated (galvanized) by the hot-dip process are required (see 3.7).
g. AQL 1.0 (see 4.4.1.1).
h. Packaging requirements (see 5.1).

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6.3 Typical bar type grating figure. Figure 1 represents a typical bar type grating for reference purposes.

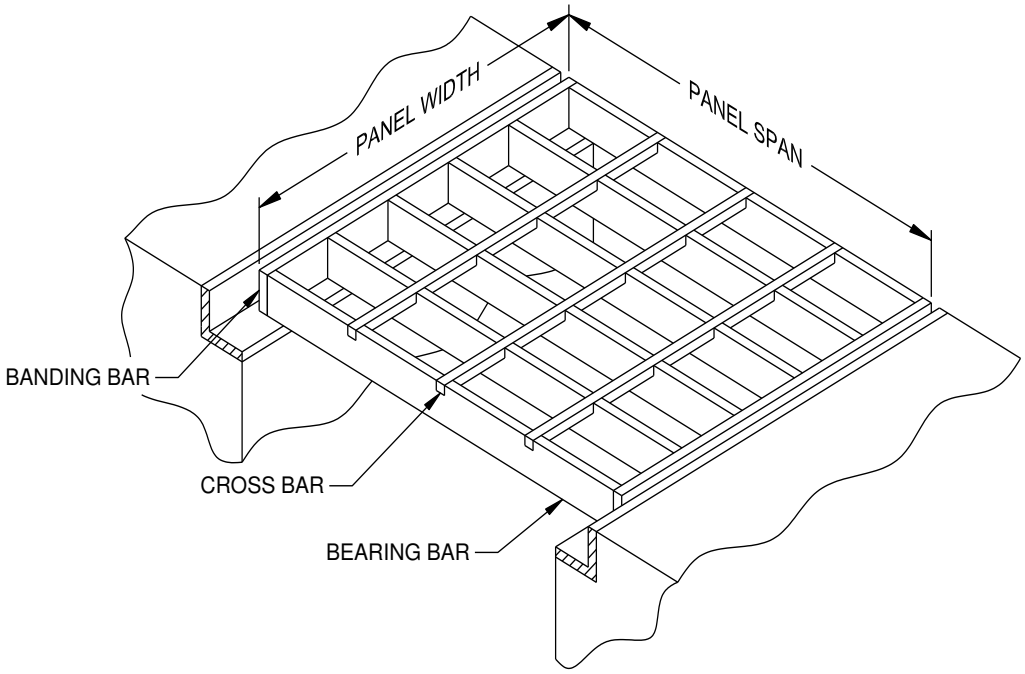


FIGURE 1. Typical grating.

6.4 Subject term (key word) listing.

Banding bar

Bearing bar

Cross bar

Reticuline

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH

Preparing activity:
Navy – SH
(Project 5670-2019-001)

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